

# DESIGN AND DEVELOPMENT OF THE BEAM COLLIMATORS FOR CiADS \*

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## Abstract

The China initiative Accelerator Driven System (CiADS) consists of a 350 m-long linac, a spallation target, a sub-critical reactor and several experimental terminals. The linac will provide protons at the energy of 500 MeV with 2.5 MW power. In order to keep the uncontrolled beam loss along the beam transport line before entering the target and the reactor less than 1 W/m, a two-stage collimation system with 2+1 periodic structure has been designed for the linac and target coupling section of CiADS. The detailed design of the beam collimation system is presented, including material selection, structural design, thermal performance analysis, radiation shielding optimization, and remote maintenance. Key technical issues which affect the collimation equipment development are also introduced.

## INTRODUCTION

In particle accelerators, beam loss management is one of the core challenges for ensuring the efficient and stable operation of the accelerators. In high-energy particle accelerators, the collimator is a key component for managing beam loss. It is mainly used for beam collimation, impurity removal and equipment protection. Its performance directly affects the operational efficiency of the accelerators, the service life of the equipment, and radiation safety [1-5]. However, the current technology still faces the following major bottlenecks: thermal damage to materials caused by high heat load and energy deposition, insufficient cooling efficiency, irradiation damage, high positioning accuracy requirements, stability and ultra-high vacuum performance requirements, as well as remote operation and maintenance of collimators during equipment failures or equipment upgrade processes. The detailed design of the beam collimation system is presented, including material selection, structural design, thermal performance analysis, radiation shielding optimization, and remote maintenance. Key technical issues which affect the collimation equipment development are also introduced.

## CONCEPTUAL DESIGN

The China initiative Accelerator Driven System (CiADS) consists of a 350 m-long linac, a spallation target, a sub-critical reactor and several experimental terminals. The linac will provide protons at the energy of 500 MeV with 2.5 MW power [6]. In order to keep the uncontrolled

beam loss along the beam transport line before entering the target and the reactor less than 1 W/m, a two-stage collimation system with 2+1 periodic structure has been designed for the linac and target coupling section of CiADS. Considering high-energy proton scattering and forward secondary particles, the first two periods include a primary collimator and a secondary collimator, while in the third period, only one secondary collimator is arranged to collect the scattered particles at the end. The beam scraping block of the primary collimator needs to be remotely synchronized for high-precision adjustment. In addition, considering the ultra-high vacuum and high stability requirements of the accelerator equipment, the main technical specifications are shown in Table 1.

Table 1: Main Specifications of the CiADS Primary Collimator

| Move-<br>ment<br>Range<br>[mm] | Adjust-<br>ment<br>precision<br>[mm] | Vacuum<br>leakage<br>rate<br>[Pa·m <sup>3</sup> ·s <sup>-1</sup> ] | Limiting<br>vacuum<br>[Pa] |
|--------------------------------|--------------------------------------|--------------------------------------------------------------------|----------------------------|
| 2~40                           | <0.1                                 | <1×10 <sup>-9</sup>                                                | <1×10 <sup>-6</sup>        |

## The Overall Scheme Design and Key Equipment Design

According to the physical requirements, the length of a single period of the first two periods is 6.16 meters, including the primary collimator, the secondary collimator, and the local shielding, as shown in Fig. 1.

According to the physical requirements, the primary collimator adopts four independently adjustable thick beam scraping blocks, with one pair in the horizontal direction and another in the vertical direction. Considering the strength of the material, thermal conductivity, beam stop ability, irradiation damage performance, etc., the material of the beam scraping block is selected as tungsten [7-8]. The size of the beam scraping block in the horizontal direction is 80 mm×50 mm×170 mm; the size of the beam scraping block in the vertical direction is 80 mm×50 mm×180 mm. Taking into account the physical properties of the tungsten block and its welding performance with oxygen-free copper, the beam scraping block is divided into multiple tungsten sheets, which are welded to the oxygen-free copper transition layer to form a tungsten-copper composite part, and then connected with other components. To ensure a good heat exchange effect, the cooling of the beam scraping block assembly is carried out by water cooling.

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The design of the beam scraping block assembly is shown in Fig. 2.

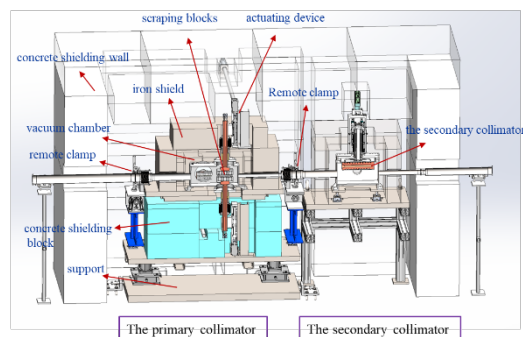


Figure 1: Schematic drawing of CiADS collimator system.

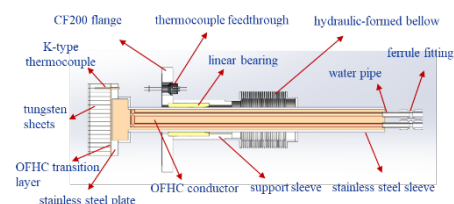


Figure 2: Schematic drawing of the beam scraping block assembly of the CiADS primary collimator.

In order to reduce the size of the shielding body and make its installation more compact, the vacuum chamber is designed as a square chamber. According to the size of the beam scraping block, the interface between the beam scraping block and the vacuum chamber is selected as the CF200 flange, which is convenient for the installation and sealing of the beam scraping block. To facilitate remote installation and disassembly, bellows are provided at the upstream and downstream of the vacuum chamber along the beam line, with the compression stroke being 30 mm. The remote clamp enables the quick disconnection and connection of the collimator with the upstream and downstream vacuum pipes.

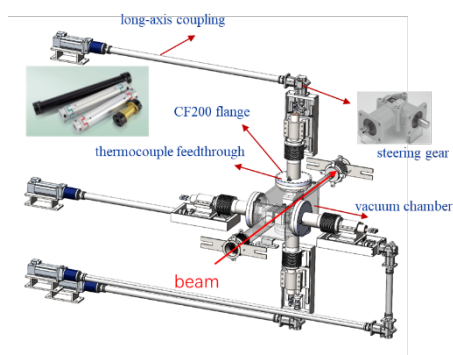


Figure 3: Schematic drawing of the primary collimator and its transmission design.

The travel range of each individual beam scraping block component is 1-20 mm, and the linear motion accuracy of each one needs to be better than 0.1 mm. The actuating device operates in a high-radiation environment, and all components should be considered for their radiation resistance performance. Considering the convenience of motor

maintenance and replacement, the motors of all actuating devices placed on the side of the aisle for easy maintenance. The primary collimator and its transmission design are shown in Fig. 3.

To prevent the equipment around the primary collimator from being damaged by high radiation and to meet the requirements for manual maintenance, it is necessary to provide good shielding for the collimator. According to the radiation protection calculation results, iron blocks, concrete blocks and concrete walls are selected from inside out to shield the radiation dose. The external part of the vacuum chamber, except for the beam outlet direction, requires a 40 cm thick iron shield, and the remaining parts need a 35 cm thick iron shield. To facilitate processing, the iron shield is divided into two layers. To facilitate the installation of the collimator and the shielding body, the support components of the collimator are divided into three layers. The bottom support and shielding steel plate are equipped with three-dimensional adjustable supports. The primary collimator weighs approximately 15 tons and can be lifted and transported as a single unit. The structure of the primary collimator is shown in Fig. 4.

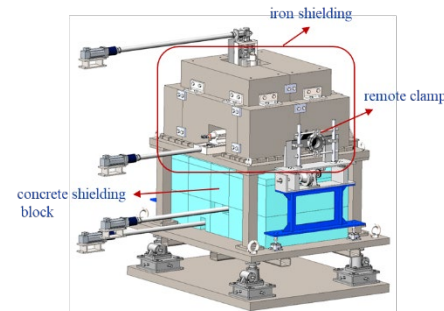


Figure 4: Schematic drawing of the primary collimator.

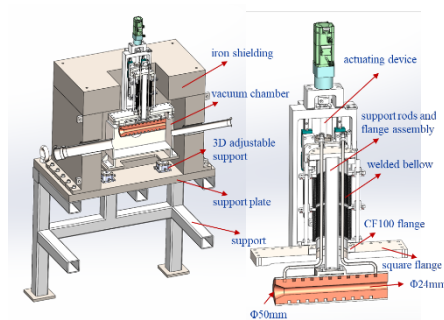


Figure 5: Schematic drawing of the secondary collimator.

The secondary collimator adopts an OFHC cylindrical beam scraping blocks and a stainless-steel sleeve. The outer diameter of the OFHC beam scraping block is 84 mm and its length is 370 mm. The center of the beam scraping block is a conical and cylindrical through hole. The outer surface of the OFHC cylinder is equipped with a spiral water channel. When the secondary collimator is not working, it needs to be lifted upwards to avoid the beam. The iron shielding body is equipped outside the vacuum chamber and pipelines, the external dimensions of the iron shielding body are: L1000×W1100×H850. Along the beam direction, the thickness of the iron shielding body is 160 mm, and the

thickness on both sides of the body is 300 mm. The total weight of the secondary collimator is 6.7 tons and it can be lifted and transported as a single unit. The structure of the secondary collimator is shown in Fig. 5.

## SIMULATION AND TEST

### *Thermomechanical Analysis of the Beam Scraping Block*

The beam scraping design was carried out with a current of 5 mA. The maximum scraping power in the first cycle was 1.93 kW. The thermal power on the primary and secondary collimators was 1.64 kW. The maximum thermal power of the single scraping block on the primary collimator was 294 W, and the maximum thermal power density was approximately 65 W/cm<sup>3</sup>. Based on this maximum scraping power, thermo-mechanical analysis calculations were conducted. When the inlet flow velocity of the cooling water was set at 5 m/s and the inlet water temperature was 27 °C, the highest temperature was located on the scraping block surface, approximately 118 °C. The temperature on the waterway wall surface was less than 33 °C. The equivalent stress at the tungsten-copper bonding surface was less than 30 MPa, which was less than one-third of the tungsten-copper welding strength (~100 MPa). The maximum deformation was less than 0.018 mm, as shown in Fig. 6, (a) is temperature distribution diagram, (b) is equivalent stress distribution diagram, (c) is deformation diagram.

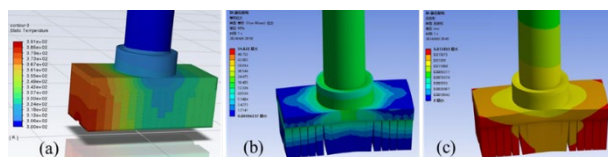


Figure 6: Schematic drawing of the secondary collimator.

### *Welding Performance Test of the Beam Scraping Block Sample and Structural Improvement*

According to the design, a set of scraper block samples was processed by brazing. The samples were subjected to high-temperature testing to verify the welding performance of the tungsten-copper composite part. The scraper block sample was placed in a vacuum furnace for high temperature testing. The test temperature was 350 °C, and it was left to stand for 2 hours before being removed. It was found that three tungsten blocks and the copper transition layer on the tungsten-copper composite part had cracks, as shown in Fig. 7(c).



Figure 7: (a) welded tungsten-copper and stainless steel composite parts, (b) the beam scraping block assembly

after brazing, (c) the beam scraping block assembly after high-temperature testing.

After high-temperature testing, cracks appeared in the tungsten-copper composite part. The main reason was the significant difference in thermodynamic performance parameters between the two materials. Therefore, to ensure the welding strength, we improved the structural design by adding a 1mm thick oxygen-free copper transition layer between the tungsten blocks before brazing the tungsten-copper composite parts. Figure 8(a) shows the welded tungsten-copper composite part. Figures 8(b) and (c) are two sections cut from the tungsten-copper composite part in Fig. 8(a). The tungsten-copper composite block in Fig. 8(b) was used to process samples for tensile and shear tests. The test results showed that the tensile strength was 360 MPa and the shear strength was 124 MPa. Figure 8(c) shows the state of the tungsten-copper composite part after being placed in a vacuum furnace for high-temperature testing at 500 °C for 2 hours. It can be seen that the tungsten-copper composite part remained intact without any cracks.

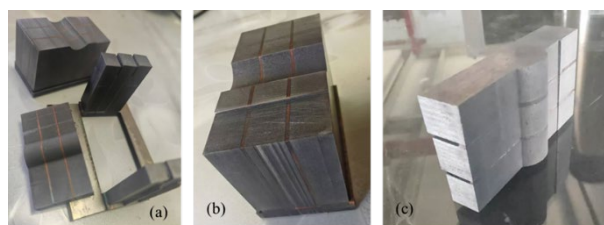


Figure 8: (a) shows the welded tungsten-copper composite part, (b) and (c) are two sections cut from the tungsten-copper composite part in Fig. 8(a).

## CONCLUSION

The detailed design of the beam collimation system of CiADS is presented, it can keep the uncontrolled beam loss along the beam transport line before entering the target and the reactor less than 1 W/m. Key technical issues which affect the collimation equipment development are also introduced, the test results show that adding an oxygen-free copper transition layer between the tungsten sheets helps to enhance the welding strength of tungsten-copper composite part, providing technical support for the development of tungsten-copper composite parts.

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